

Metric Spaces: In functional analysis we shall study more general “spaces” and “functions” defined on them. A metric space is a pair (X, d) , where X is a set and d is a metric on X (or distance function on X), that is, a function defined on $X \times X$ such that for all $x, y, z \in X$ we have following four properties.

- 1st Property: M_1 d is real – valued, finite and non – negative
 2nd Property: M_2 $d(x, y) = 0$ if and only if $x = y$
 3rd Property: M_3 $d(x, y) = d(y, x)$ (Symmetry)
 4th Property: M_4 $d(x, y) \leq d(x, z) + d(z, y)$ (Triangle Inequality)

The above four properties called axioms of metric space. As metric space is ordered pair so we take $X \times X$ mean two elements from set X .

Equality (3 points on same line): $d(x, y) = d(x, z) + d(z, y)$

Inequality (3 points of triangle): $d(x, y) \leq d(x, z) + d(z, y)$

Now if we have more than three points say $x_1, x_2, x_3, \dots, x_n$ then distance between any two point say x_1 and x_n is $d(x_1, x_n) \leq d(x_1, x_2) + d(x_2, x_3) + d(x_3, x_4) + \dots + d(x_{n-1}, x_n)$ is the generalized triangle inequality.

Subspace: A subspace (Y, d) of (X, d) is obtained if we take a subset $Y \subset X$ and restrict d to $Y \times Y$. Thus the metric on Y is the restriction $\tilde{d} = d|_{Y \times Y}$ $\tilde{d}$ is called the metric induced on Y by d .

Example of Metric Space:

- Real line $\mathbb{R}$
- Euclidean plane $\mathbb{R}^2$
- 3-Dimensional Euclidean Space $\mathbb{R}^3$
- nTuples Euclidean Space $\mathbb{R}^n$
- Unitary Space $\mathbb{C}^n$
- Complex Plane $\mathbb{C}$
- Sequence Space ℓ^∞
- Functional Space $C[a, b]$
- Discrete Metric Space
- Sequence Space S
- Space ℓ^p
- Hilbert Sequence ℓ^2

Examples of Metric Space:

Real line $\mathbb{R}$ forms a metric Space

Let x and y be two real points on real line, then

$$d(x, y) = |x - y| \quad ; \quad x, y \in \mathbb{R}$$

Now we prove all the four properties (axioms) of metric space.

$$d(x, y) = |x - y|$$

$$d(x, y) = |x - z + z - y| \quad ; \quad z \in \mathbb{R}$$

$$d(x, y) \leq |x - z| + |z - y|$$

$$= d(x, z) + d(z, y)$$

Euclidean plane $\mathbb{R}^2$ forms a metric Space

Euclidean space mean that the points are taken from $\mathbb{R}^2$ in ordered pair.

$$\mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}$$

Suppose that one point is (x_1, y_1) and the other point is (x_2, y_2) , then the distance d between these two points is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Thus $(\mathbb{R}^2, d)$ is a metric space

Sequence Space ℓ^∞

If we take all sequences of complex number which are bounded in a set then the set is called ℓ^∞ . The distance function is $d(x, y) = \sup_{i \in \mathbb{N}} |\xi_i - \eta_i|$

This sequence of complex numbers space, $d(x, y)$ form a metric space.

Function Space $C[a, b]$

Here $C[a, b]$ is a function space and x and y are functions on t variable and also real valued (its value is always real value) as

$$x: x(t) \in \mathbb{R} \quad \text{and} \quad y: y(t) \in \mathbb{R} \quad (\text{Real values and continuous}).$$

Discrete Metric Space

$$\begin{aligned} d(x, y) &= 0 && \text{if } x \text{ and } y \text{ are same} \\ \text{and } d(x, y) &= 1 && \text{if } x \text{ and } y \text{ are different.} \end{aligned}$$

Sequence Space s:

Here the distance function is changed from previous one, the metric d defined by

$$d(x, y) = \sum_{j=1}^{\infty} \frac{1}{2^j} \frac{|\xi_j - \eta_j|}{1 + |\xi_j - \eta_j|}$$

where $x = (\xi_j)$ and $y = (\eta_j)$ and are all complex nos.

$$\begin{aligned} \frac{|\xi_i - \alpha_i + \alpha_i - \eta_i|}{1 + |\xi_i - \alpha_i + \alpha_i - \eta_i|} &\leq \frac{|\xi_i - \alpha_i|}{1 + |\xi_i - \alpha_i|} + \frac{|\alpha_i - \eta_i|}{1 + |\alpha_i - \eta_i|} \\ \Rightarrow \frac{|\xi_i - \eta_i|}{1 + |\xi_i - \eta_i|} &\leq \frac{|\xi_i - \alpha_i|}{1 + |\xi_i - \alpha_i|} + \frac{|\alpha_i - \eta_i|}{1 + |\alpha_i - \eta_i|} \\ \sum \frac{1}{2^i} \cdot \frac{|\xi_i - \eta_i|}{1 + |\xi_i - \eta_i|} &\leq \sum \frac{1}{2^i} \cdot \frac{|\xi_i - \alpha_i|}{1 + |\xi_i - \alpha_i|} + \sum \frac{1}{2^i} \cdot \frac{|\alpha_i - \eta_i|}{1 + |\alpha_i - \eta_i|} \end{aligned}$$

Space l^p

$$d(x, y) = \left(\sum_{j=1}^{\infty} |\xi_j - \eta_j|^p \right)^{\frac{1}{p}}$$

where $y = (\eta_j)$ and $\sum_{j=1}^{\infty} |\eta_j|^p < \infty$

The real Space l^p If the elements ξ_j and η_j are not complex numbers but from real numbers then the space is called real space l^p .

The complex Space l^p If the elements ξ_j and η_j are complex numbers then the space is called complex space l^p

The Hilbert Sequence Space l^2 Now the case $p=2$ (fixed) The Hilbert sequence space l^2 with the metric defined by

$$\begin{aligned} d(x, y) &= \left(\sum_{j=1}^{\infty} |\xi_j - \eta_j|^2 \right)^{\frac{1}{2}} \\ &= \sqrt{\sum_{j=1}^{\infty} |\xi_j - \eta_j|^2} \end{aligned}$$

Open Set and Closed Set on Real Line $\mathbb{R}$

On real line we have open set not open ball. (2,5) is an open set. It includes all values between 2 and 5 but does not include 2 and 5. [2,5] is a closed set. It includes all values between 2 and 5 including 2 and 5.

Open Set and Closed Set on Real Line $\mathbb{R}^2$

In $\mathbb{R}^2$ we have open ball. Here open ball has center x_0 and radius r. It includes all values but does not include boundaries

Points on the Boundary:

$d(x_0, x) = r$; x is on boundary.

All the points are lies on the boundary. If the difference of that point x from the center x_0 is r. $d(x_0, x) = r$ x is a boundary point.

Points inside the Boundary

All the points are lies inside the boundary if the distance between x and the center x_0 (i-e. difference of that point x from the center x_0) is less than r. $d(x_0, x) < r$ x lies inside the boundary.

Points outside the Boundary

All the points are lies outside the boundary if the distance between x and the center x_0 (i-e. the difference of that point x from the center x_0) is greater than r. $d(x_0, x) > r$ x lies outside the boundary.

Open Ball and Closed Ball in $\mathbb{R}^2$

Open Ball: In $\mathbb{R}^2$ if boundary is not included then it is open ball. It means all points inside the boundary are included. $\Rightarrow d(x_0, x) < r$

Closed Ball: In $\mathbb{R}^2$ if boundary is included then it is closed ball. It means all points inside the boundary and on the boundary are included. $\Rightarrow d(x_0, x) \leq r$

Open Sphere, Closed Sphere in $\mathbb{R}^3$

In $\mathbb{R}^3$ we have open sphere, closed sphere.

Ball and Sphere (General Form)

Open Ball: $B(x_0; r) = \{x \in X \mid d(x, x_0) < r\}$

Closed Ball: $\tilde{B}(x_0; r) = \{x \in X \mid d(x, x_0) \leq r\}$

Sphere: $S(x_0; r) = \{x \in X \mid d(x, x_0) = r\}$

Sphere: Sphere includes all those points which are exactly lies on the boundary or on the radius r. It does not have any inside or outside points.

In discrete space, we have defined distance function. Sphere can be empty.

$$d(x, x) = 0$$

$$d(x, y) = 1 \quad ; \quad x \neq y$$

$$d(x, x_c) < r$$

$$d(x, x_c) \leq r$$

$$\text{Sphere} = \emptyset$$

For boundary we can subtract open ball from closed ball. $S(x_0, r) = \tilde{B}(x_0, r) - B(x_0, r)$

Open Set: A subset M of a metric space X is said to be open if it contains a ball about each of its points.

Closed Set: A subset K of X is said to be closed if *its complement (in X) is open*.
that is $K^c = X - K$ is open.

Neighborhood of a Point: We can find an open ball around each point in Open set.

“An open ball $B(x_0, \varepsilon)$ of radius ε is often called an ε -neighborhood of x_0 .” By a neighborhood of x_0 , we mean any subset of X which contains an ε -neighborhood of x_0

Difference between Radius r and ε For radius r means larger values, 0.1, 0.5, 10, 40 while radius ε means very small values like 0.002, 0.000003 etc.

Interior Point: We call x_0 an interior point of a set $M \subset X$ if M is a neighborhood of x_0 . The interior of M is the set of all interior points of M. **Int(M)** is open and is the largest open set contained in M. Collection of all open balls is an open set whether the radius ε of open ball is greater or smaller.

Topological Space: Let $\mathfrak{T}$ be collection of all open subsets of X. Then $(X, \mathfrak{T})$ is said to be a topological space if it satisfies following properties:

T1): $\emptyset \in \mathfrak{T}$ and $X \in \mathfrak{T}$

T2): The union of any member of $\mathfrak{T}$ is a member of $\mathfrak{T}$.

T3): The intersection of finitely many members of $\mathfrak{T}$ is a member of $\mathfrak{T}$.

Continuous Mappings: Let $X = (X, d)$ and $Y = (Y, \tilde{d})$ be metric spaces.

A mapping $T : X \rightarrow Y$ is said to be continuous at a point $x_0 \in X$

if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $\tilde{d}(Tx, Tx_0) < \varepsilon$ for all x satisfying $d(x, x_0) < \delta$

Continuous Mapping: A mapping T of a metric space X into a metric space Y is continuous if and only if the inverse image of any open subset of Y is an open subset of X. Conversely we consider inverse image of open subset is open and prove that it is continuous.

Proof: Suppose a mapping $T : X \rightarrow Y$ and T is continuous.

Now we will prove that inverse image of any open subset in Y is open in X.

Let $S \subset Y$ be open subset. Let S_0 be the inverse image of S.

$$\text{Space } X \rightarrow \text{space } Y$$

We have to prove that this S_0 open. For this we have two cases.

1st case:

Suppose that we have chosen the element has no inverse image then

$$S_0 = \emptyset \Rightarrow \text{open} \quad (\text{because empty set is always open})$$

2nd case:

Now suppose S_0 is not empty then there is at least one

Point x_0 such that

$$S_0 \neq \emptyset \Rightarrow x_0 \in S_0$$

$$y_o = Tx_o$$

S is open , there exist an ε neighborhood N of y_o

since T is continuous $\exists S_o$ neighborhood of x_o which is mapped into N .

Since for $N \subset S$ we have $N_o \subset S_o$ so,

S_o is open on $x_o \in S_o$ and has δ neighborhood.

Accumulation Point (limit point): If M is a subset of a metric space X then x_o is a limit point of M . if it is the limit of an eventually non-constant sequence (a_i) of points of M (or limit point of M) if every neighborhood of x_o contains at least one point $y \in M$ distinct from x_o .

Translation in the form of metric space:

Let M be a subset of a metric space X , then a point x_o of X (which may or may not be a point of M) is called an accumulation point of M (or limit point of M) if every neighborhood of x_o contains at least one point $y \in M$ distinct from x_o .

Example (1) $\mathbb{R} : d(x, y) = |x - y|$

$$M(0,1)$$

$0 \notin M = (0,1)$ is a limit point of M .

1 is also a limit point of M .

as
$$\lim_{x \rightarrow \infty} \frac{1}{n} = 0, \quad 1 + \frac{1}{n}$$

Another example:

The set of integers has no limits points, $\mathbb{Z} \subset \mathbb{R}$ has no limit point, e.g. any sequence in $\mathbb{Z}$ converging to any integer is eventually constant.

Closure of M : The set consisting of the points of M and the accumulation points of M is called the closure of M and is denoted by $\bar{M}$.

Dense Set: A subset M of a metric space X is said to be dense in X if $\bar{M} = X$

Closure set is a set along with its limits points. **Example :** The rational numbers $\mathbb{Q}$ are dense in $\mathbb{R}$ is also present in $\mathbb{R}$.

Separable Spaces: A metric space X is said to be separable if it has a countable subset which is dense in X . It has two conditions; First its subset is dense and second is countable.

Examples of Separable Spaces

1. The Real Line $\mathbb{R}$
2. The Complex Plane $\mathbb{C}$
3. Discrete Metric Space
4. Space l^p

1. $\mathbb{Q}$ is dense in $\mathbb{R}$ (means $\bar{\mathbb{Q}} = \mathbb{R}$) and $\mathbb{Q}$ is countable. Hence $\mathbb{R}$ is a separable.
 $\mathbb{Q}^n$ is a countable subset of $\mathbb{R}^n$ which is dense $\mathbb{R}^n$, so $\mathbb{R}^n$ is separable space.

2. $\mathbb{C}$ has a subset $\mathbb{Q}^2$ which is dense and countable so, $\mathbb{C}$ is separable.

3. Discrete Metric Space: In discrete metric space if elements are same then distance is 0 and if elements are different then distance is 1. There is no condition on set. Set can be any set.

4. Space l^p

Space mean "a set", in this set elements are sequences, which may be real (called real space l^p) or complex numbers (called complex space $\sum_{n=1}^{\infty} x_n$). A subset of $\sum_{n=1}^{\infty} x_n$ is dense and also countable then $\sum_{n=1}^{\infty} x_n$ is a separable space.

Bounded Sequence: We call a nonempty subset $M \subset X$ a bounded set if its diameter $\delta(M) = \sup_{x, y \in M} d(x, y)$ is finite. A sequence (x_n) in X is bounded sequence if the corresponding point set is a bounded subset of X . Hence bounded sequence means finite diameter and if diameter is infinite then sequence is unbounded.

Convergence of a sequence: A sequence (x_n) in a metric space $X = (X, d)$ is said to converge or to be convergent if there is an $x \in X$ such that $\lim_{n \rightarrow \infty} d(x_n, x) = 0$ x is called the limit of (x_n) and we write

$$\lim_{n \rightarrow \infty} x_n = x \text{ or simply } x_n \rightarrow x$$

$$x_n = \frac{1}{n},$$

Example 1: n varies as $\{1, 2, 3, \dots\}$

$$\left\{ \frac{1}{n} \right\}_{n=1}^{\infty} \rightarrow 0$$

Its domain is set of natural numbers. if n varies from 1 to ∞ then $\frac{1}{n}$ approaches to 0. Or its limit point x is 0.

Example 2: $x_n = (-1)^n$ here x varies from 1 to n .

1 n -even not convergent

-1 n -odd convergent.

As this sequence is not converging at one value. It varies between 1 and -1 so it is not converging.

Example 3: $x_n = \begin{cases} 1 & \text{if } n \text{ is a square} \\ 0 & \text{if otherwise} \end{cases}$ are not convergent.

Convergence of a sequence: (Another Definition)

We say that (x_n) converges to x or has the limit x , if (x_n) is not convergent, it is said to be divergent.

$$(x_n) ; x_n \rightarrow x$$

Lemma: Let $X = (X, d)$ be a metric space, then

a) A convergent sequence in X is bounded and its limit is unique.

b) If $x_n \rightarrow x$ and $y_n \rightarrow y$ in X then $d(x_n, y_n) \rightarrow d(x, y)$

a):

Given that sequence is convergent and we have to prove that it is also bounded.

Bounded means that its corresponding diameter is finite.

For convergent sequence mean for every $\varepsilon > 0$ there exist $N = N(\varepsilon)$ such that for all x_n with $n > N$ lie in the ε -neighborhood $B(x; \varepsilon)$ of x .

As it is true for all ε so we choose

$\varepsilon = 1$, then we will find there exist N such that

$$d(x_n, x) < 1$$

We have values $x_1, x_2, \dots, x_N, x_{N+1}, \dots$

which are entries of sequence.

If we choose $n < N$, say from this part of the sequence $x_1, x_2, \dots, x_N$ then $d(x_n, x)$ is greater than 1, and

if we choose $n > N$, say from this part of the sequence $x_{N+1}, \dots$ then $d(x_n, x) < 1$,

$$d(x_n, x) < 1 \quad \forall n > N$$

Now we have calculated the distance of x from the point x_i where $i = 1, 2, \dots, N$ is

$$d(x_1, x), d(x_2, x), \dots, d(x_N, x)$$

We take the maximum of all these distances, $\max(\dots)$,

let say this max distance is " d ".

Now the distance before N is less than d and the value after N is less than 1,

$$d(x_n, x) < d + 1, \quad \forall n$$

In first part (a) we have to prove two things

i): sequence is convergent (i.e we have to prove that it is bounded)

and

ii): converging value is unique.

2nd Part Uniqueness:

Let say that $x_n \rightarrow x$, $x_n \rightarrow z$

$$d(x, z) \geq 0 \dots \dots \dots (i)$$

If we take any two values from a set then the distance between them is always greater than 0.

For uniqueness we have to prove that $x = z$, in other word the distance between x and z is $d(x, z) = 0$

Using the 4th axioms of metric space that

$$d(x, z) \leq d(x, x_n) + d(x_n, z)$$

x_n converges to x and also z (our supposition),

$$d(x, x_n) \rightarrow 0$$

and $d(x_n, z) \rightarrow 0$
 now $d(x, z) \leq 0 \dots\dots\dots(ii)$
 From (i and (ii)
 $d(x, z) = 0$
 $\Rightarrow x = z$

Hence proved that it converges to a unique value.

Cauchy Sequence: A sequence (x_n) in a metric space $X = (X, d)$ is said to be Cauchy (or fundamental) if for every $\varepsilon > 0$ there is an $N = N(\varepsilon)$ such that $d(x_m, x_n) < \varepsilon$ for every $m, n > N$.

Equivalent notation $d(x_m, x_n) \rightarrow 0$ as $m, n \rightarrow \infty$

Completeness: The space X is said to be complete if every Cauchy sequence in X converges that is, has a limit which is an element of X .

Theorem Convergent Sequence: Every convergent sequence in a metric space is a Cauchy sequence.

Proof:

Let $\{x_n\}$ be a convergent sequence such that $x_n \rightarrow x$ for every $\varepsilon > 0$ there exist $N = N(\varepsilon)$ such that $d(x_n, x) < \frac{\varepsilon}{2} \quad \forall n > N$

Now we have to prove that $\{x_n\}$ is a Cauchy sequence, for this we have to prove that

$$d(x_m, x_n) < \varepsilon ; \quad m, n > N$$

We first choose that $m > N$ then by triangular inequality,

$$d(x_m, x_n) \leq d(x_m, x) + d(x_n, x) ; \quad m, n > N$$

as $d(x_m, x) < \frac{\varepsilon}{2} , \quad d(x_n, x) < \frac{\varepsilon}{2}$

$$\Rightarrow d(x_m, x_n) \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} ; \quad m, n > N$$

$$\Rightarrow d(x_m, x_n) < \varepsilon ; \quad m, n > N$$

That we have to prove, hence $\{x_n\}$ is a Cauchy sequence.

Converse:

Now we check that every Cauchy sequence (x_n) in that space is convergent.

The converse is not true i.e. "every Cauchy sequence (x_n) in that space is not convergent".

Example 1:

The counter example is

$$a_n = \left\{ \frac{1}{n} \right\}_{n=1}^{\infty}$$

This is a Cauchy sequence in $(0, 1]$ but it is not convergent in $(0, 1]$.

Closure is a collection of limit points and the set itself.

Limit point is such point that if we draw an open ball around it then we can find another point other than that point which belongs to that set.

A set is called a **closed set** if all the limits points are present in that set.

Theorem:

Let M be a nonempty subset of a metric space $d(X, d)$ and $\bar{M}$ its closure as defined before then,

a): $x \in \bar{M}$ if and only if there is a sequence (x_n) in M such that $x_n \rightarrow x$.

b): M is closed if and only if the situation $x_n \in M, \quad x_n \rightarrow x$ implies that $x \in M$.

a): Proof:

$$x \in \bar{M} \Leftrightarrow \text{there is a sequence } (x_n) \text{ in } M \text{ such that } x_n \rightarrow x$$

$\bar{M}$ is a collection of M and its limit points. Now there are two options,

i): x belong to $M, \quad x \in \bar{M}$

ii): x does not belong to $M, \quad x \notin M$

i) $x \in \bar{M}$

now if x does not belong to M then x is a limit point of M and if $x \in M$ then $x_1 = x, \quad x_2 = x, \quad x_3 = x, \dots\dots\dots$ or $(x_1, x_2, \dots\dots\dots) = (x_n) = (x_1 x_2 \dots\dots\dots),$

hence $x_n \rightarrow x \in M$

ii) $x \notin M$

For every $n=1,2,3,\dots$ the ball $B\left(x; \frac{1}{n}\right)$, here ε is $\frac{1}{n}$.

Containing an $x_n \in M$, other than x .

Now as

$$n \rightarrow \infty \Rightarrow \frac{1}{n} \rightarrow 0 \Rightarrow x_n \rightarrow x.$$

Hence x_n converges to x .

b):

M is closed, if and only if the situation $x_n \in M, x_n \rightarrow x$.

$$\Rightarrow x \in M$$

M is closed if and only if $\bar{M} = M$,

Now we have to prove that $\bar{M} = M$ for this we prove that $\bar{M} \subseteq M$, $M \subseteq \bar{M}$

i): $\bar{M} \subseteq M$

by definition M contains M and its limit point so this condition is fulfilled.

ii): $M \subseteq \bar{M}$

Now we prove that $M \subseteq \bar{M}$

Let $x \in \bar{M}$, we will show that $x \in M$.

Now if we take x belongs to $\bar{M}$ then from above "a" part of this theorem, we have a sequence x_n in M such that $x_n \rightarrow x$ this implies $x \in M$.

That means $M \subseteq \bar{M}$.

Hence $\bar{M} = M$

Theorem (Continuous Mapping): A mapping $T: X \rightarrow Y$ of a metric space (X, d) into a metric space $(Y, \tilde{d})$ is continuous at a point $x_0 \in X$ if and only if $x_n \rightarrow x_0$ implies $Tx_n \rightarrow Tx_0$

Proof:

Suppose T is continuous, we will prove that if $x_n \rightarrow x_0$ implies $Tx_n \rightarrow Tx_0$.

T is continuous means $T: X \rightarrow Y$

a given $\varepsilon > 0$ there exist $\delta > 0$ such that

$$d(x, x_0) < \delta \Rightarrow \tilde{d}(Tx, Tx_0) < \varepsilon$$

So, let $x_n \rightarrow x_0$ there exist a $\mathbb{N}$ such that for all $n > \mathbb{N}$ we have

$$d(x_n, x_0) < \delta$$

This is δ of convergence.

$$\tilde{d}(Tx_n, Tx_0) < \varepsilon, \quad n > \mathbb{N}$$

By definition $Tx_n \rightarrow Tx_0$

Examples (Completeness):

$\mathbb{R}$ is a complete metric space which means every sequence in $\mathbb{R}$ is convergent in $\mathbb{R}$ and every Cauchy sequence is convergent.

- Every Cauchy sequence in a metric space is bounded. This is for every metric space.
- If a Cauchy sequence has a subsequence that converges to $\bar{x}$, then the sequence converges to $\bar{x}$.
- Every sequence of real numbers has a monotone subsequence.

Theorem:

$\mathbb{R}$ is a complete metric space, i.e., every Cauchy sequence of real numbers converges.

Proof:

Let $\{x_n\}$ be a Cauchy sequence. It implies that $\{x_n\}$ is bounded. Now if the given Cauchy sequence is bounded then its subsequence is also bounded.

Every subsequence of $\{x_n\}$ is bounded. Also $\{x_n\}$ has a monotone subsequence. Now $\{x_n\}$ is monotone as well as bounded.

Monotone Convergence Theorem:

If a sequence $\{x_n\}$ is monotone and bounded this implies that it is convergent.

EXAMPLES OF COMPLETENESS

- The Euclidean space $\mathbb{R}^n$ is complete.
- $\mathbb{C}[a, b]$ is complete metric space
- l^∞ is complete metric space
- Space $\mathbb{Q}$
- Space of Polynomials
- Isometric mappings/spaces

Isometric Mappings: Let $X = (X, d)$ and $\tilde{X} = (\tilde{X}, d)$ be metric spaces. A mapping $T: X \rightarrow \tilde{X}$ is said to be isometric or isometry if T preserve distance. Preseve distance mean after applying the mapping the distance is preserve, i.e. for all $x, y \in X$

$$\tilde{d}(T_x, T_y) = d(x, y)$$

Isometric Spaces: The space X is said to be isometric with space $\tilde{X}$ if there exist a bijective isometry of X onto $\tilde{X}$. X and $\tilde{X}$ are then called isometric spaces.

Theorem(Completion)

For a metric space $X = (X, d)$ there exists a complete metric space $\hat{X} = (\hat{X}, d)$ which has a subspace W that is isometric with X and is dense in $\hat{X}$.

Vector Space: A vector space (or linear space) over a field K is a nonempty set X of elements $x, y, \dots$ (called vectors) together with two algebraic operations. These operations are called vector addition and multiplication of vectors by scalars, that is, by elements of K.

Examples of Vector Space

- Space $\mathbb{R}^n$
- Space $\mathbb{C}^n$
- Space $\mathbb{C}[a, b]$
- Space l^2

1. Space $\mathbb{R}^n$

$$x = (\xi_1, \dots, \xi_n), \quad \xi_i \in \mathbb{R} \quad y = (\eta_1, \dots, \eta_n), \quad \eta_i \in \mathbb{R}$$

Addition:

$$x + y = (\xi_1 + \eta_1, \dots, \xi_n + \eta_n)$$

scalar Multiplication:

let α be a scalar then

$$\alpha x = (\alpha \xi_1, \dots, \alpha \xi_n)$$

Now addition and scalar multiplication in $\mathbb{R}^n$ is a vector space.

2. Space $\mathbb{C}^n$

Addition:

$$\text{Let } x = (\xi_1, \dots, \xi_n), \quad \xi_i \in \mathbb{C} \\ y = (\eta_1, \dots, \eta_n), \quad \eta_i \in \mathbb{C}$$

Scalar Multiplication:

addition and scalar multiplication is same as in $\mathbb{R}^n$, so $\mathbb{C}^n$ is a vector space.

3. Space $\mathbb{C}[a, b]$

$$\text{Let } x \in \mathbb{C}[a, b] \quad \text{and} \quad y \in \mathbb{C}[a, b]$$

where x and y are fucntions and operating on t

Addition:

$$(x + y)(t) = x(t) + y(t)$$

Scalar Multiplication:

$$(\alpha x)(t) = \alpha x(t)$$

So under addition and scalar multiplication $\mathbb{C}[a, b]$ is vector space over a field $\mathbb{R}$ or $\mathbb{C}$.

4. Space l^2 :

In this space we have sequences, if $x \in l^2$ then x is a sequence, say

$$x = (\xi_1, \dots, \xi_n), \quad x \in l^2$$

$$\text{and } y = (\eta_1, \dots, \eta_n), \quad y \in l^2$$

Addition:

$$x + y = (\xi_1 + \eta_1, \dots, \xi_n + \eta_n)$$

Scalar Multiplication:

$$\alpha x = (\alpha \xi_1, \dots, \alpha \xi_n)$$

So under addition and scalar multiplication the space l^2 is vector space over a field $\mathbb{R}$ or $\mathbb{C}$

Vector Space (Subspace): A subspace of a vector space X is a nonempty subset Y of X such that addition and scalar multiplication are closed in Y .

Two Types of subspaces

- Improper Subspace: If the span of a subspace is equal to that vector space ;
- Proper Subspace: If the span of a subspace is not equal to that vector space

Linear Combination A linear combination of vectors $x_1, \dots, x_n$ of a vector space X is an expression of the form $a_1x_1 + \dots + a_nx_n$ where the coefficients $a_1, \dots, a_n$ are any scalars.

Span of a Set:

For any nonempty subset $M \subset X$ the set of all linear combinations of vectors of M is called the span of M . Written as “span M ”.

Linear Independence:

If two vectors have same direction and different in magnitude then one vector is multiple of other which means that one is dependent to other.

Basis of a Vector Space: As span of M is also a subspace, if the subspace (collection of vectors) is improper subspace (means span of M is equal to that vector space) and linearly independent (coefficients are equal to zero) then that particular subspace is a Basis of a Vector Space. So, for basis the subspace have to improper subspace and linear independent.

Vector Space Dimension (definition):

The number of elements in subspace of a basis is called dimension of that vector space.

Dimension

i. Finite dimensional vector space

A vector space which is spanned by a finite set of vectors is said to be finite dimensional vector space.

Examples: In $\mathbb{R}^n$ space, Elements of basis of $\mathbb{R}^n$ are $e_1, e_2, \dots, e_n$,

$e_1 = (1, 0, \dots, 0), e_2 = (0, 1, \dots, 0), \dots, e_n = (0, 0, \dots, 1)$ Sometimes it is called Canonical basis of $\mathbb{R}^n$ basis $\mathbb{R}^n$. Similarly in $\mathbb{C}^n$ space n -dimension

ii. Infinite dimensional vector space

A vector space which cannot be spanned by a finite set of vectors is said to be infinite dimensional vector space. **Examples:** $C[a, b]$ is infinite dimension vector space because there is no finite set which can span the set of function. In l^2 space, there are sequences, this is also infinite dimensional vector space.

- Every nonempty vector space $X \neq \{0\}$ has a basis.

Theorem:

Let X be an n dimensional vector space. Then any proper subspace Y of X has dimension less than n .

Proof:

If $n=0$ this implies $X=\{0\}$

There is no proper subspace. Hence we can't continue.

If dimension of Y is zero.

$$\dim Y=0$$

and

$$X \neq Y \quad Y=\{0\}$$

$$\{Y \text{ is proper subspace of } X\}$$

$$\dim Y < \dim X$$

suppose $\dim Y=n$

⇒ Y would have a basis of n elements.

⇒ that basis would also be a basis for X , as element in basis are same, they span and linearly independent.

$$\dim X=n \quad \text{when basis are same then } X=Y$$

but it is contradict to our supposition as we suppose that Y is a proper subset of X i.e $Y \subset X$ which means X and Y are not equal.

⇒ any linearly independent set of vectors in Y must have less elements than n .

⇒ $\dim Y < n$

That we have to prove.

Norm (definition): A norm on a (real or complex) vector space X is a real-valued function on X whose value at an $x \in X$ is denoted by $\|x\|$.

$$\text{i):} \quad \|x\| \geq 0 \quad (\text{N1})$$

$$\text{ii):} \quad \|x\| = 0 \quad \Leftrightarrow \quad x = 0 \quad (\text{N2})$$

$$\text{iii): } \|\alpha x\| = |\alpha| \|x\| \quad (\text{N3})$$

$$\text{iv): } \|x + y\| \leq \|x\| + \|y\| \quad (\text{N4}) \quad \text{triangular inequality}$$

Norm metric: A norm on X defines a metric d on X which is given by

$$d(x, y) = \|x - y\| \quad \text{where } x, y \in X$$

and is called the metric induced by the norm as this metric depend on norm so we call it metric induced by norm.

Norm Space: A normed space X is a vector space with a norm defined on it. A normed space is denoted by $(X, \|\cdot\|)$ or simply by X .

Banach Space: A Banach space is a complete normed space, (Complete in the metric defined by the norm).

Examples of Normed Space

➤ **Euclidean Space** $\mathbb{R}^n$

➤ **Unitary Space** $\mathbb{C}^n$

➤ **Space** l^p

➤ **Space** l^∞

➤ **Space** $\mathbb{C}[a, b]$

Euclidean Space $\mathbb{R}^n$

This is a metric space and elements in $\mathbb{R}^n$ is in n-tuples form,

$$x = (\xi_1, \xi_2, \dots, \xi_n) \quad \text{where } \xi_i \in \mathbb{R}, \quad x \in X$$

$$\|x\| = \sqrt{|\xi_1|^2 + \dots + |\xi_n|^2}$$

$$= \left(\sum_{i=1}^n |\xi_i|^2 \right)^{\frac{1}{2}}$$

$$y = (\eta_1, \eta_2, \dots, \eta_n) \quad \text{where } \eta_i \in \mathbb{R}$$

The distance function $d(x, y) = \|x - y\|$

$$d(x, y) = \sqrt{|\xi_1 - \eta_1|^2 + \dots + |\xi_n - \eta_n|^2}$$

Unitary Space $\mathbb{C}^n$

This is a metric space and elements in $\mathbb{C}^n$ is in n-tuples form,

$$x = (\xi_1, \xi_2, \dots, \xi_n) \quad \text{where } \xi_i \in \mathbb{C}, \quad x \in X$$

$$\|x\| = \sqrt{|\xi_1|^2 + \dots + |\xi_n|^2}$$

$$= \left(\sum_{i=1}^n |\xi_i|^2 \right)^{\frac{1}{2}}$$

$$y = (\eta_1, \eta_2, \dots, \eta_n) \quad \text{where } \eta_i \in \mathbb{C}$$

The distance function

$$d(x, y) = \|x - y\|$$

$$= \sqrt{|\xi_1 - \eta_1|^2 + \dots + |\xi_n - \eta_n|^2}$$

Space l^p

$$x = (\xi_1, \xi_2, \dots) , \quad y = (\eta_1, \eta_2, \dots)$$

$$\|x\| = \left(\sum_{j=1}^{\infty} |\xi_j|^p \right)^{\frac{1}{p}}$$

The distance function

$$d(x, y) = \|x - y\|$$

$$= \left(\sum_{j=1}^{\infty} |\xi_j - \eta_j|^p \right)^{\frac{1}{p}}$$

Space l^∞

$$x \in l^\infty$$

The metric is given by

$$\|x\| = \sup_j |\xi_j|$$

Space $C[a, b]$:

This is a space of all real valued continuous functions defined on closed interval $[a, b]$
The norm of the function is $\|x\| = \max_{t \in J} |x(t)|$, with this metric space it is a norm space.

Unit Sphere: The sphere with center 0 and radius 1, $S(0;1)$, this we define in $\mathbb{R}^2$, but in any metric space. Those points from x whose norm is 1. $\{x \in X \mid \|x\| = 1\}$,

In a normed space X is called the unit sphere. In norm space the collection of all those points which are equal to 1 is called a Unit Sphere.

Normed Spaces Subspace (definition) A subspace Y of a normed space X is a subspace of X considered as a vector space, with the norm obtained by restricting the norm on X to the subset Y . This norm on Y is said to be induced by the norm on X . If Y is closed in X , then Y is called a closed subspace of X .

Subspace l^p :

A subspace Y of a Banach space X is a subspace of X considered as a normed space. Hence we do not require Y to be complete.

Theorem :

A subspace Y of a Banach space X is complete if and only if the set Y is closed in X .

Convergence in Normed Spaces.

The metric function is $d(x, y) = \|x - y\|$

For convergence we define as

i): A sequence (x_n) in a normed space X is convergent if X contains an x such that

$$\lim_{n \rightarrow \infty} \|x_n - x\| = 0$$

$$x_n \rightarrow x, \quad x \text{ limit of } (x_n)$$

Now this definition define for Cauchy sequence

ii): A sequence (x_n) in a normed space X is a Cauchy sequence if for every $\varepsilon > 0$ there is an N such that

$$\|x_m - x_n\| < \varepsilon \quad \text{for all } m, n > N$$

Convergence of Infinite Series

If s_n convergent, $s_n \rightarrow s$, then

$$\sum_{i=1}^{\infty} x_i = x_1 + x_2 + \dots \text{ is also convergent.}$$

if $\|s_n - s\| \rightarrow 0$ then $s_n \rightarrow s$.

If we have following series

$$\|x_1\| + \|x_2\| + \dots \text{ converges,}$$

$$\Rightarrow \sum_{i=1}^{\infty} x_i \text{ absolutely convergent.}$$

Basis: In a normed space X is a Cauchy sequence if for every $\varepsilon > 0$ there is an N such that Elements are spanning and are linearly independent.

Any element $x = \alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_n e_n$ in the form of norm is

$$\|x - \alpha_1 e_1 - \alpha_2 e_2 - \dots - \alpha_n e_n\| \rightarrow 0$$

and if this condition is hold then we say that it is a basis in the norm space.

Theorem Completion: Let $X = (x, \|\cdot\|)$ be a normed space then there is a Banach space $\hat{X}$ and an isometry

A from X onto a subspace W of $\hat{X}$ which is dense in $\hat{X}$. The space $\hat{X}$ is unique, except for isometries.

Lemma (Linear Combination): Let $\{x_1, \dots, x_n\}$ be a linearly independent set of vectors in a normed space X (of any dimension). Then there is a number $c > 0$ such that for every choice of scalars $\alpha_1, \dots, \alpha_n$ we have $\|\alpha_1 x_1 + \dots + \alpha_n x_n\| \geq c(|\alpha_1| + \dots + |\alpha_n|)$

Normed Spaces Completeness

Every finite dimensional subspace Y of a normed space X is complete. In particular, every finite dimensional normed space is complete.

Normed Spaces Closedness

As we have already proved that every finite dimensional subspace is complete and we also know that a subspace is complete if and only if it is closed.

THEOREM: Every finite dimensional subspace Y of a normed space X is closed in X . This result is true for finite dimensional subspace but for infinite space it is not true.

Normed Spaces Equivalent Norms

A norm $\|\cdot\|$ on a vector space X is said to be equivalent to a norm $\|\cdot\|_0$ on X if there are positive numbers a and b such that for all $x \in X$ we have

$$a\|x\|_0 \leq \|x\| \leq b\|x\|_0$$

Theorem One finite dimensional vector space X , any norm $\|\cdot\|$ is equivalent to any other norm $\|\cdot\|_0$

Compactness:

A metric space X is said to be compact if every sequence in X has a convergent subsequence. A subset M of X is said to be compact if M is compact considered as a subspace of X , that is if every sequence in M has a convergent subsequence whose limit is an element of M .

Lemma (Compactness)

A compact subset M of a metric space is closed and bounded.

Proof:

For close of M we show that $\bar{M} = M$. Now we have to prove closed and bounded.

1st It is closed:

We will show that $\bar{M} = M$.

We know a conclusion $M \subset \bar{M}$ now we will prove its converse i.e. $\bar{M} \subset M$

By definition for every $x \in \bar{M}$ there exist a sequence (x_n) such that x_n approaches to x .

Now it is given that M is compact i.e contains limit of every convergent subsequence.

It implies that $x \in M$

It implies $M = \bar{M}$

Hence M is closed.

2nd It is bounded:

To prove bounded, suppose on contrary let it is not bounded. It implies it would contain unbounded sequence (y_n) such that

$D(y_n, b) > n$, which is any fixed element.

But this sequence could not have a convergent subsequence since a convergent subsequence must be bounded. It is a contradiction. It implies M is bounded.

Conversely

In general the converse of this lemma is false. The converse is false that a closed and bounded metric space is not compact.

Lemma (Compactness) In a finite dimensional normed space X , any subset $M \subset X$ is compact if and only if M is closed and bounded.

Proof:

Let M be closed and bounded, $\dim X = n$ and $\{e_1 + \dots + e_n\}$ be a basis for X , $\langle x_m \rangle$ be any sequence in

$$M \Rightarrow x_m = \xi_1^{(m)} e_1 + \dots + \xi_n^{(m)} e_n$$

Now we Consider any sequence $\langle x_m \rangle$ in M . Each x_m has a representation

$$x_m = \xi_1^{(m)} e_1 + \dots + \xi_n^{(m)} e_n \quad \text{since } C > 0$$

Hence the sequence of numbers $\xi_j^{(m)}$ is bounded and by Bolzano Weierstrass theorem, has a point of accumulation $\xi_j^{(m)}$ $1 \leq j \leq n$ as we conclude that x_m has a subsequence z_m which converges to $z = \sum \xi_j e_j$.

Since M is closed $z \in M$. This shows that arbitrary sequence x_m in M has a subsequence which converges in M . Hence M is compact.

F. Riesz's Lemma

Let Y and Z be subspaces of a normed space X (of any dimension), and suppose that Y is closed and is a proper subset of Z , then for every real number θ in the interval $(0,1)$ there is a $z \in Z$ such that

$$\|z\| = 1$$

$$\|z - y\| \geq \theta \quad \text{for all } y \in Y$$

Theorem (Finite Dimension) If a normed space X has the property that the closed unit ball $M = \{x \mid \|x\| \leq 1\}$ is compact, then X is finite dimensional.

Theorem (Continuous Mapping) Let X and Y be metric spaces and $T : X \rightarrow Y$ be a continuous mapping. Then the image of a compact subset M of X under T is compact.

Proof:

By definition of compactness we need to show that every sequence $\langle y_n \rangle$ in the image $T(M) \subset Y$ continuous a subsequence which converges in $T(M)$.

Now since $y_n \in T(M)$, we have x_n such that $y_n = Tx_n$, for some $x_n \in M$. since M is compact, (x_n) contains subsequence $\langle x_{n_k} \rangle$ which converges in M .

The image of (x_{n_k}) is a subspace of (y_n) which converge in $T(M)$.
It implies that $T(M)$ is compact.

Theorem: maximum and minimum

A continuous mapping T of a compact subset M of a metric space X into $\mathbb{R}$ assumes a maximum and a minimum at some points of M .

$$T : M \rightarrow \mathbb{R}$$

$$\Rightarrow T(M) \subset \mathbb{R}$$

$$\left. \begin{array}{l} T(M), \\ M - \text{compact} \\ T - \text{continuous} \end{array} \right\} \text{by previous result}$$

$$\Rightarrow T(M) \text{ is compact.}$$

which means it is closed and bounded because compactness implies close and bounded.

$$\Rightarrow \inf T(M) \in T(M), \quad \text{and} \quad \sup T(M) \in T(M)$$

Inverse image of these two points consist of points of M at which Tx is minimum or maximum respectively. And that we have to prove.

Operator : Operator is a mapping whose domain and range is a vector space. It is subset of vector space.

Linear Operator: A linear operator T is an operator such that

- i): the domain $D(T)$ of T is a vector space and the range $R(T)$ lies in a vector space over the same field.
- ii): for all $x, y \in D(T)$ and scalar α

$$T(x+y) = Tx + Ty \quad \text{also} \quad T(\alpha x) = \alpha Tx$$

By combining above two equations

$$T(\alpha x + \beta y) = \alpha Tx + \beta Ty \quad \text{where } \alpha \text{ and } \beta \text{ are both scalar}$$

$T(x) = Tx$ is same.

Some more notations.

$D(T)$ domain of T

$R(T)$ range of T

$N(T)$ denotes the null space of T .

Null space are those element from the domain of T such that on which we operate gives the answer zero.
 $x \in D(T)$ such that $Tx=0$. Also null space of T is similar to kernel of T . T is a homomorphism when it is a linear operator.

Linear Operators Examples

Identity Operator

Identity mean it operate on the same vector space. $I_x : X \rightarrow X$

$$\Rightarrow I_x(x) = x \quad \forall x \in X$$

$$\Rightarrow I_x(\alpha x + \beta y) \quad \text{we have to prove}$$

Zero Operator:

$$O : X \rightarrow Y \text{ such that } OX = 0 \quad \forall x \in X$$

here the 0 on right side is belong to vector space Y .

Differentiation:

Let X be a vector space of all polynomials on $[a, b]$. A set of polynomial in denoted by $x(t)$

$$Tx(t) = x'(t) \quad \forall x(t) \in X$$

When we apply T on polynomial $x(t)$ then $x'(t)$ is also a polynomial. So this operator T maps X onto itself.
There is no polynomial whose derivative we can't find.

Integration:

Linear operator T for $C[a, b]$ into itself can be defined by

$$Tx(t) = \int_a^t x(\tau) d\tau$$

taa τ is just a variable and $C[a, b]$ is collection of all continuous function on a and b .

Multiplication by t:

Let $C[a, b]$ be a collection of continuous functions defined on a and b .

$$Tx(t) = tx(t)$$

This operator plays an important role in quantum theory of physics.

Elementary vector algebra:

Matrices: We denote matrix by capital letter say A. whose elements are in rows and column. $A = (\alpha_{jk})$

For matrix multiplication number of first matrix column is equal to number of rows of second column.

Range: Null space is the collection of those elements from the domain on which we apply the operator and the answer is zero.

Theorem Let T be a linear operator. Then:

- The range $R(T)$ is a vector space. (domain is also a vector space as discussed)
- If $\dim D(T) = n < \infty$, then $\dim R(T) \leq n$ (dimension of domain vector space is finite then range is equal or less than the dimension of domain or equal.
- The null space $N(T)$ is a vector space.

The first two results are about range and third result is about null space.

Proof: (a) $R(T)$ is a vector space.

$$y_1, y_2 \in R(T)$$

$$\Rightarrow \alpha y_1 + \beta y_2 \in R(T), \text{ where } \alpha, \beta \text{ are scalar}$$

Since

$$y_1, y_2 \in R(T) \text{ and } x_1, x_2 \in D(T)$$

$$T : D(T) \rightarrow Y$$

$$y_1 \in Tx_1, y_2 \in Tx_2$$

Also domain of T " $D(T)$ " is a vector space so, $\alpha x_1 + \beta x_2 \in D(T)$ this is by definition of vector space. Since T is linear

$$T(\alpha x_1 + \beta x_2) = \alpha Tx_1 + \beta Tx_2 = \alpha y_1 + \beta y_2 \in R(T)$$

Here $\alpha x_1 + \beta x_2$ is domain and gives $\alpha y_1 + \beta y_2$ range of T . Hence $R(T)$ is a vector space.

Part (c):

$$x_1, x_2 \in N(T)$$

$$Tx_1 = Tx_2 = 0$$

To prove it a vector space, we have to prove $\alpha x_1 + \beta x_2 \in N(T)$

$$T(\alpha x_1 + \beta x_2) = \alpha Tx_1 + \beta Tx_2 = \alpha \times 0 + \beta \times 0 = 0$$

$$\Rightarrow \alpha x_1 + \beta x_2 \in N(T)$$

$$\Rightarrow N(T) \text{ is a vector space} \quad (\text{proved})$$

Inverse Operators: Operator is a mapping whose domain and range is vector space. Particular in norm space, there is also inverse mapping. For inverse operator the same condition is one-to-one and onto. One-to-one means image of different elements is different. And onto means the range covers all the set of domain. If these two conditions hold then we can define inverse operator.

$T : D(T) \rightarrow Y$ is said to be injective or one-to-one if for any

$$x_1, x_2 \in D(T) \text{ such that } x_1 \neq x_2 \Rightarrow Tx_1 \neq Tx_2$$

$$T^*Tx = x \quad \forall x \in D(T) \quad \text{and} \quad TT'y = y \quad \forall y \in R(T)$$

Inverse exist if and only if null space has only zero. There is only zero in null space

Theorem (Inverse Operator) Let X, Y be vector spaces, both real or both complex. Let $T : D(T) \rightarrow Y$ be a linear operator with domain $D(T) \subset X$ and range $R(T) \subset Y$. then:

a): The inverse $T' : R(T) \rightarrow D(T)$ exists if and only if $Tx=0 \Rightarrow x=0$. (i.e null space has zero elements).

b): If T' exists, it is a linear operator.

c): if $\dim D(T) = n < \infty$ and T^{-1} exists, then $\dim R(T) = \dim D(T)$.

as there is if and only if condition so we have to prove in both ways.

Inverse of Product: Bijective mean one to one and onto. Here it means inverse of T and S exists. Let $T : X \rightarrow Y$ and $S : Y \rightarrow Z$ be bijective linear operators, where X, Y are vector spaces.

Then the inverse $(ST)^{-1} : Z \rightarrow X$ of the product (the composite) ST exists, and $(ST)^{-1} = T^{-1}S^{-1}$.

Proof: If S is bijective and T is bijective then ST is also bijective.

$$ST : X \rightarrow Z \text{ bijective} \Rightarrow (ST)^{-1} \text{ exist.}$$

It means if $(ST)(ST)^{-1} = I_Z$

If $S : Y \rightarrow Z$ then $S^{-1}S = I_Y$

$$S^{-1}ST(ST)^{-1} = S^{-1}I_Z \Rightarrow T(ST)^{-1} = S^{-1} \Rightarrow T^{-1}T(ST)^{-1} = T^{-1}S^{-1} \Rightarrow$$

$$(ST)^{-1} = T^{-1}S^{-1}$$